

Proof. 1. **Notation.** Let

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n,$$

and set

$$A = \sum_{i=1}^n a_i^2, \quad B = \sum_{i=1}^n b_i^2, \quad C = \sum_{i=1}^n a_i b_i.$$

2. **Degenerate case.** If $B = 0$ then every $b_i = 0$; consequently $C = 0$ and $AB = 0$. Hence

$$\left(\sum_{i=1}^n a_i b_i \right)^2 = 0 = AB,$$

so the inequality holds with equality. In the sequel we assume $B > 0$.

3. **Quadratic construction.** For a real parameter t consider

$$Q(t) = \sum_{i=1}^n (a_i - t b_i)^2.$$

Each term is a square, therefore $Q(t) \geq 0$ for all $t \in \mathbb{R}$.

4. **Expansion.** Expanding gives

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 - 2t a_i b_i + t^2 b_i^2) \\ &= A - 2Ct + Bt^2. \end{aligned}$$

Thus $Q(t) = Bt^2 - 2Ct + A$ is a quadratic polynomial with leading coefficient $B > 0$.

5. **Discriminant argument.** A quadratic $Bt^2 - 2Ct + A$ that is non-negative for every real t must have non-positive discriminant:

$$\Delta = (-2C)^2 - 4BA = 4(C^2 - AB) \leq 0.$$

Hence $C^2 \leq AB$.

6. **Desired inequality.** Since $C = \sum_{i=1}^n a_i b_i$, $A = \sum_{i=1}^n a_i^2$ and $B = \sum_{i=1}^n b_i^2$, the inequality $C^2 \leq AB$ is exactly

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

7. **Equality condition.** Equality occurs iff $\Delta = 0$, i.e. $C^2 = AB$. With $B > 0$ this means the quadratic $Q(t)$ has a double root, so

$$t_0 = \frac{C}{B}$$

satisfies $a_i = t_0 b_i$ for every i ; that is, $\mathbf{a}$ is a scalar multiple of $\mathbf{b}$. In the degenerate case $B = 0$ we already have equality. Hence equality holds precisely when $\mathbf{b} = 0$ or $\mathbf{a}$ and $\mathbf{b}$ are linearly dependent (one is a scalar multiple of the other).

8. **Conclusion.** For any real sequences $(a_i)_{i=1}^n$ and $(b_i)_{i=1}^n$ we have

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right),$$

with equality exactly in the cases described above. $\square$

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